SepsisDx Classification Report

Su et al. 2021 — Simulated Patient Data from Urinary Biomarker Panel Study
Generated March 06, 2026 at 15:43 | Model: Random Forest (Kaggle-trained, threshold=0.3323)

Source Data

Su L, et al. *Sci Rep.* 2021;11:20794. 151 simulated patients (53 Sepsis, 48 SIRS, 50 Healthy). CRP/Cr mapped from µg/mmol to pg/mg. IP-10/Cr simulated.

Classification Summary

Group	N	Correct	Accuracy	Avg P(sep)
Sepsis	53	38/53	71.7%	0.666
SIRS	48	46/48	95.8%	0.074
Healthy	50	50/50	100.0%	0.021
Overall	151	134/151	88.7%	—

Sensitivity: 0.717 **Specificity:** 0.980 **PPV:** 0.950 **NPV:** 0.865 **TP:** 38 **TN:** 96 **FP:** 2 **FN:** 15

SEPSIS Group — 38/53 Correctly Classified (71.7%)

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
1	Sepsis_037	1010.7	192.3	Septic	0.9968	VERY HIGH	Septic
2	Sepsis_047	903.0	144.1	Septic	0.9966	VERY HIGH	Septic
3	Sepsis_025	906.8	155.2	Septic	0.9916	VERY HIGH	Septic
4	Sepsis_023	1329.4	167.3	Septic	0.9913	VERY HIGH	Septic
5	Sepsis_045	2180.1	187.5	Septic	0.9911	VERY HIGH	Septic
6	Sepsis_034	2660.9	119.0	Septic	0.9902	VERY HIGH	Septic
7	Sepsis_038	2568.1	211.8	Septic	0.9901	VERY HIGH	Septic
8	Sepsis_049	536.8	91.1	Septic	0.9882	VERY HIGH	Septic
9	Sepsis_029	1420.7	316.7	Septic	0.9868	VERY HIGH	Septic
10	Sepsis_019	3722.2	98.5	Septic	0.9867	VERY HIGH	Septic
11	Sepsis_017	936.4	99.9	Septic	0.9866	VERY HIGH	Septic
12	Sepsis_035	655.6	136.3	Septic	0.9843	VERY HIGH	Septic
13	Sepsis_042	593.8	123.9	Septic	0.9836	VERY HIGH	Septic
14	Sepsis_005	766.1	97.9	Septic	0.9801	VERY HIGH	Septic
15	Sepsis_020	1724.7	207.7	Septic	0.9798	VERY HIGH	Septic
16	Sepsis_015	4148.0	186.1	Septic	0.9762	VERY HIGH	Septic
17	Sepsis_051	438.1	287.2	Septic	0.9675	VERY HIGH	Septic

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
18	Sepsis_044	1159.1	150.9	Septic	0.9666	VERY HIGH	Septic
19	Sepsis_018	869.0	73.9	Septic	0.9664	VERY HIGH	Septic
20	Sepsis_030	2508.4	69.0	Septic	0.9657	VERY HIGH	Septic
21	Sepsis_006	1143.1	221.3	Septic	0.9616	VERY HIGH	Septic
22	Sepsis_007	657.3	73.8	Septic	0.9601	VERY HIGH	Septic
23	Sepsis_003	1040.2	78.2	Septic	0.9441	VERY HIGH	Septic
24	Sepsis_002	1708.2	68.2	Septic	0.9423	VERY HIGH	Septic
25	Sepsis_048	241.8	72.8	Septic	0.9412	VERY HIGH	Not Septic
26	Sepsis_032	235.2	57.6	Septic	0.8552	VERY HIGH	Not Septic
27	Sepsis_014	2233.1	51.0	Septic	0.8078	VERY HIGH	Septic
28	Sepsis_052	908.2	62.1	Septic	0.8055	VERY HIGH	Septic
29	Sepsis_026	818.5	58.9	Septic	0.8046	VERY HIGH	Septic
30	Sepsis_033	610.4	63.8	Septic	0.7916	HIGH	Septic
31	Sepsis_024	1883.7	46.7	Septic	0.7306	HIGH	Septic
32	Sepsis_021	1000.9	56.3	Septic	0.6894	HIGH	Septic
33	Sepsis_039	3221.2	44.8	Septic	0.6530	HIGH	Septic
34	Sepsis_043	376.2	54.6	Septic	0.4346	MODERATE	Septic
35	Sepsis_022	230.7	47.0	Septic	0.4294	MODERATE	Not Septic
36	Sepsis_010	339.7	38.5	Septic	0.4173	MODERATE	Septic
37	Sepsis_046	1208.0	37.0	Septic	0.3801	MODERATE	Septic
38	Sepsis_053	1528.0	33.8	Septic	0.3577	MODERATE	Septic
39	Sepsis_012	1776.6	35.4	Not Septic ✗	0.3201	MODERATE	Septic
40	Sepsis_004	261.1	30.5	Not Septic ✗	0.3133	MODERATE	Not Septic

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
41	Sepsis_027	454.4	30.4	Not Septic ✗	0.3084	MODERATE	Septic
42	Sepsis_040	491.3	30.7	Not Septic ✗	0.2825	MODERATE	Septic
43	Sepsis_013	1711.0	29.4	Not Septic ✗	0.2818	MODERATE	Septic
44	Sepsis_016	1395.0	27.8	Not Septic ✗	0.2604	MODERATE	Septic
45	Sepsis_031	815.9	30.0	Not Septic ✗	0.2203	MODERATE	Septic
46	Sepsis_011	509.9	19.6	Not Septic ✗	0.2037	MODERATE	Septic
47	Sepsis_028	593.4	20.8	Not Septic ✗	0.1906	LOW	Septic
48	Sepsis_041	468.6	20.1	Not Septic ✗	0.1847	LOW	Septic
49	Sepsis_050	1714.3	12.8	Not Septic ✗	0.0537	LOW	Septic
50	Sepsis_008	425.7	14.2	Not Septic ✗	0.0389	LOW	Septic
51	Sepsis_009	935.4	11.2	Not Septic ✗	0.0319	LOW	Septic
52	Sepsis_036	1126.1	6.0	Not Septic ✗	0.0242	LOW	Septic
53	Sepsis_001	922.2	8.8	Not Septic ✗	0.0098	LOW	Septic

SIRS Group — 46/48 Correctly Classified (95.8%)

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
1	SIRS_033	291.6	40.0	Septic ✗	0.5011	HIGH	Not Septic
2	SIRS_034	789.8	53.7	Septic ✗	0.3930	MODERATE	Septic
3	SIRS_018	233.9	30.4	Not Septic	0.2905	MODERATE	Not Septic
4	SIRS_021	423.8	31.9	Not Septic	0.2731	MODERATE	Septic
5	SIRS_013	3600.4	34.0	Not Septic	0.2453	MODERATE	Septic
6	SIRS_043	339.0	26.0	Not Septic	0.2270	MODERATE	Septic
7	SIRS_032	1450.7	22.6	Not Septic	0.2225	MODERATE	Septic
8	SIRS_003	365.6	22.6	Not Septic	0.1598	LOW	Septic
9	SIRS_027	155.8	19.5	Not Septic	0.1467	LOW	Not Septic
10	SIRS_005	424.2	16.5	Not Septic	0.1403	LOW	Septic
11	SIRS_019	224.7	22.5	Not Septic	0.1353	LOW	Not Septic
12	SIRS_035	125.7	16.2	Not Septic	0.0812	LOW	Not Septic
13	SIRS_020	423.2	15.4	Not Septic	0.0642	LOW	Septic
14	SIRS_011	678.0	13.5	Not Septic	0.0468	LOW	Septic
15	SIRS_038	453.1	14.0	Not Septic	0.0366	LOW	Septic
16	SIRS_012	349.3	10.3	Not Septic	0.0332	LOW	Septic
17	SIRS_031	137.0	14.7	Not Septic	0.0314	LOW	Not Septic
18	SIRS_006	330.6	13.6	Not Septic	0.0275	LOW	Septic
19	SIRS_029	1710.5	-12.1	Not Septic	0.0255	LOW	Septic
20	SIRS_044	380.7	12.4	Not Septic	0.0245	LOW	Septic
21	SIRS_016	244.7	12.7	Not Septic	0.0233	LOW	Not Septic
22	SIRS_009	2418.3	11.1	Not Septic	0.0219	LOW	Septic
23	SIRS_037	346.1	-22.0	Not Septic	0.0214	LOW	Septic
24	SIRS_017	112.4	13.3	Not Septic	0.0207	LOW	Not Septic
25	SIRS_002	357.2	4.9	Not Septic	0.0203	LOW	Septic

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
26	SIRS_023	1101.6	0.4	**Not Septic**	0.0192	**LOW**	Septic
27	SIRS_042	365.3	7.9	**Not Septic**	0.0183	**LOW**	Septic
28	SIRS_015	184.8	-5.4	**Not Septic**	0.0176	**LOW**	Not Septic
29	SIRS_022	2943.8	-7.8	**Not Septic**	0.0172	**LOW**	Septic
30	SIRS_026	619.3	-10.7	**Not Septic**	0.0171	**LOW**	Septic
31	SIRS_007	2050.0	1.1	**Not Septic**	0.0164	**LOW**	Septic
32	SIRS_030	313.8	-5.2	**Not Septic**	0.0164	**LOW**	Septic
33	SIRS_008	1245.2	-11.7	**Not Septic**	0.0162	**LOW**	Septic
34	SIRS_001	388.5	-4.3	**Not Septic**	0.0162	**LOW**	Septic
35	SIRS_028	2197.8	6.7	**Not Septic**	0.0158	**LOW**	Septic
36	SIRS_040	668.6	3.8	**Not Septic**	0.0139	**LOW**	Septic
37	SIRS_048	1406.1	2.7	**Not Septic**	0.0138	**LOW**	Septic
38	SIRS_039	241.1	-2.2	**Not Septic**	0.0136	**LOW**	Not Septic
39	SIRS_041	159.1	0.8	**Not Septic**	0.0135	**LOW**	Not Septic
40	SIRS_046	153.7	-5.5	**Not Septic**	0.0135	**LOW**	Not Septic
41	SIRS_036	222.8	-4.8	**Not Septic**	0.0134	**LOW**	Not Septic
42	SIRS_024	211.1	-3.9	**Not Septic**	0.0134	**LOW**	Not Septic
43	SIRS_014	220.1	6.4	**Not Septic**	0.0134	**LOW**	Not Septic
44	SIRS_010	173.8	2.1	**Not Septic**	0.0126	**LOW**	Not Septic
45	SIRS_004	178.1	-15.9	**Not Septic**	0.0126	**LOW**	Not Septic
46	SIRS_025	174.8	4.2	**Not Septic**	0.0126	**LOW**	Not Septic
47	SIRS_045	698.6	7.6	**Not Septic**	0.0125	**LOW**	Septic
48	SIRS_047	129.3	-15.1	**Not Septic**	0.0100	**LOW**	Not Septic

HEALTHY Group — 50/50 Correctly Classified (100.0%)

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
1	Healthy_002	100.7	17.7	Not Septic	0.1093	LOW	Not Septic
2	Healthy_009	99.2	21.3	Not Septic	0.1088	LOW	Not Septic
3	Healthy_037	99.7	17.2	Not Septic	0.1043	LOW	Not Septic
4	Healthy_036	99.8	16.5	Not Septic	0.0990	LOW	Not Septic
5	Healthy_023	100.5	15.8	Not Septic	0.0853	LOW	Not Septic
6	Healthy_032	99.8	15.4	Not Septic	0.0529	LOW	Not Septic
7	Healthy_040	100.0	14.6	Not Septic	0.0307	LOW	Not Septic
8	Healthy_027	99.8	13.1	Not Septic	0.0207	LOW	Not Septic
9	Healthy_048	99.8	14.1	Not Septic	0.0207	LOW	Not Septic
10	Healthy_004	99.5	9.8	Not Septic	0.0177	LOW	Not Septic
11	Healthy_030	101.7	11.1	Not Septic	0.0159	LOW	Not Septic
12	Healthy_045	99.3	-0.1	Not Septic	0.0100	LOW	Not Septic
13	Healthy_035	100.0	-13.7	Not Septic	0.0100	LOW	Not Septic
14	Healthy_049	99.1	0.9	Not Septic	0.0100	LOW	Not Septic
15	Healthy_031	99.6	4.3	Not Septic	0.0100	LOW	Not Septic
16	Healthy_033	101.3	1.8	Not Septic	0.0100	LOW	Not Septic
17	Healthy_034	101.1	-12.2	Not Septic	0.0100	LOW	Not Septic

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
18	Healthy_047	99.8	3.4	Not Septic	0.0100	LOW	Not Septic
19	Healthy_044	99.7	-3.9	Not Septic	0.0100	LOW	Not Septic
20	Healthy_046	100.0	-1.6	Not Septic	0.0100	LOW	Not Septic
21	Healthy_038	101.4	-1.8	Not Septic	0.0100	LOW	Not Septic
22	Healthy_039	100.5	-3.8	Not Septic	0.0100	LOW	Not Septic
23	Healthy_029	99.9	-4.1	Not Septic	0.0100	LOW	Not Septic
24	Healthy_041	102.4	-11.1	Not Septic	0.0100	LOW	Not Septic
25	Healthy_042	99.6	-8.9	Not Septic	0.0100	LOW	Not Septic
26	Healthy_043	100.7	6.4	Not Septic	0.0100	LOW	Not Septic
27	Healthy_001	100.4	-0.5	Not Septic	0.0100	LOW	Not Septic
28	Healthy_026	99.2	2.9	Not Septic	0.0100	LOW	Not Septic
29	Healthy_028	100.5	2.2	Not Septic	0.0100	LOW	Not Septic
30	Healthy_025	101.7	-1.2	Not Septic	0.0100	LOW	Not Septic
31	Healthy_003	101.3	1.3	Not Septic	0.0100	LOW	Not Septic
32	Healthy_005	99.6	-5.0	Not Septic	0.0100	LOW	Not Septic
33	Healthy_006	99.8	-11.8	Not Septic	0.0100	LOW	Not Septic
34	Healthy_007	99.9	-9.7	Not Septic	0.0100	LOW	Not Septic
35	Healthy_008	100.3	-7.3	Not Septic	0.0100	LOW	Not Septic
36	Healthy_010	100.0	-8.2	Not Septic	0.0100	LOW	Not Septic
37	Healthy_011	99.8	8.4	Not Septic	0.0100	LOW	Not Septic
38	Healthy_012	100.4	-9.0	Not Septic	0.0100	LOW	Not Septic
39	Healthy_013	99.2	9.3	Not Septic	0.0100	LOW	Not Septic
40	Healthy_014	100.5	3.8	Not Septic	0.0100	LOW	Not Septic

#	ID	CRP/Cr	IP-10/Cr	Prediction	P(Sep)	Risk	SepsisDx
41	Healthy_015	101.4	-1.6	**Not Septic**	0.0100	**LOW**	Not Septic
42	Healthy_016	99.3	-0.4	**Not Septic**	0.0100	**LOW**	Not Septic
43	Healthy_017	99.3	-6.5	**Not Septic**	0.0100	**LOW**	Not Septic
44	Healthy_018	99.4	4.5	**Not Septic**	0.0100	**LOW**	Not Septic
45	Healthy_019	99.2	-4.5	**Not Septic**	0.0100	**LOW**	Not Septic
46	Healthy_020	99.9	-12.3	**Not Septic**	0.0100	**LOW**	Not Septic
47	Healthy_021	99.9	-12.8	**Not Septic**	0.0100	**LOW**	Not Septic
48	Healthy_022	101.3	1.7	**Not Septic**	0.0100	**LOW**	Not Septic
49	Healthy_024	100.1	1.6	**Not Septic**	0.0100	**LOW**	Not Septic
50	Healthy_050	100.3	6.4	**Not Septic**	0.0100	**LOW**	Not Septic

Disclaimer: Simulated data from published summary statistics. IP-10/Cr simulated. For algorithm validation only.